

Bram J. Banik

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Education

BASc in Engineering Physics - University of British Columbia

Expected April 2028

- Dean's List
- 85.3%
- Primary Courses of Interest: ENPH253 (Intro to Instrument Design), MATH255 (Differential Equations), ELEC202 (Circuit Analysis II), MATH217 (Multivariable/Vector Calculus), ENPH259 (Experimental Techniques), ENPH353 (Engineering Physics Project I)

Westview High School - Beaverton, OR, USA | GPA: 4.143 | ACT: 36

September 2019 - April 2023

Skills

Electrical: KiCad, PCB Design, Soldering, Filters and Noise Reduction, Debugging

Mechanical: Solidworks, Solidworks Simulation, OnShape, ANSYS, CNC Mill, Laser and Waterjet Cutting, Carbon Fiber Wet Layup and Vacuum Infusion

Software: C, C++, Java, MATLAB, Python, HTML + CSS, Linux, ESP32 freeRTOS, Machine Learning

General: GitHub, Waveforms, Excel, Adobe Illustrator, Adobe After Effects, Notion, Obsidian

Experience

UBC Thunderbikes Aerodynamics Team Lead

August 2025 - Present

- Lead subteam of 10+ students for UBC Thunderbikes first foray into the MotoStudent competition
- Heading projects such as carbon-fiber fairings and custom composite rear frame design
- Managing \$4000 budget for the aerodynamics and composites subteam
- Outlined training for CAD, CNC, and aerodynamics simulations for all-electric racing motorcycle

Engineering Physics Student Life Coordinator

July 2025 - Present

- Organizing events for the Engineering Physics student body
- Managing a budget of \$1000+ to distribute among both semesters
- Reigniting interest in social events, with 30+ students attending

UBC Thunderbikes Aerodynamics Subteam Member

September 2024 - July 2025

- Designed and built a carbon fiber gas tank and numerous battery panels for an all-electric motorcycle
- Hands-on experience with wet-layup and vacuum infusion processing techniques

Projects

Fully Autonomous Pet-Rescuing Robot Competition

- Built a robot that successfully drove, detected, and picked up a stuffed pet autonomously
- Utilized custom computer vision model to detect pets, communicating between a Raspberry Pi and an ESP32
- Designed and soldered PCBs for all circuits on the robot, including dual H-Bridges and switch debouncing
- Wrote object-oriented codebase in C++ using freeRTOS to allow efficient time-stitching
- Debugged and handled noise issues from drive motors using shielding and software-based solutions

SIFT Homography Tracker

- Developed a GUI used QtDesigner to display live webcam tracking footage
- Used OpenCV's Python library to detect object via the SIFT algorithm and FLANN keypoint matching
- Coded a homography detector when significant keypoints were present using RANSAC

Analog Motor Control Circuit

- Built and tested a circuit to control the speed of a motor via a potentiometer
- Used circuit analysis techniques and Waveforms to debug circuit
- Implemented photodiode output and DAC to count motor RPM, using Op-Amps to amplify signal
- Wrote detailed lab reports to track bugs and circuit improvements, peer reviewed other circuit designs